

# Lecture (14): Thalamus, Limbic System & Hypothalamus

(Dr. Wagdy Nashaat Habeeb)

## 1. Thalamus

- “The highest subcortical sensory center for all sensations except olfaction”
- Called “The secretary of CC”

➤ **Site:** At the lat. walls of the 3<sup>rd</sup> ventricle (one on each side)

➤ Thalamic nuclei & their functions:

(I) Anatomical classification: (according to their situation)

5 groups; ant., post., med., lat. & ventral groups

(II) Physiological classification: (according to their function)

(A) Non-specific projection nuclei:

= Midline & intralaminar nuclei:

- Receive signals from: RAS

- Project to: All areas of CC

- Functions:

1. Alertness & wakefulness → Facilitation of CC

2. Perceptions of protopathic sensations

(B) Specific projection nuclei (Cortical relay nuclei):

### 1. PLVNT<sup>1</sup> & PMVNT<sup>2</sup>:

- Receive signals from: Spinal, med.<sup>1</sup> & trigeminal<sup>2</sup> lemnisci
- Project to: Somatic sensory areas of CC

#### 2. Function: Relay station for epicritic sensations

MGB<sup>1</sup> & LGB<sup>2</sup>:

- Receive signals from: Auditory<sup>1</sup> & visual<sup>2</sup> pathways
- Project to: <sup>1</sup>Auditory & <sup>2</sup>visual cortex
- Function: Relay station for <sup>1</sup>auditory & <sup>2</sup>visual sensations

#### (C) Others:

### 1. VL nucleus:

- Receive signals from: Ipsilateral BG & contralateral neoCB
- Project to: Motor cortex
- Function: The only thalamic motor nucleus that connects BG & CB with motor cortex

### 2. Ant. nucleus:

- Receive signals from: Hypothalamus (mammillary body)
- Project to: Limbic lobe
- Function: Recent memory & emotions

### 3. Dorsomedial<sup>1</sup> & dorsolateral<sup>2</sup> nuclei:

- Receive signals from: Other thalamic nuclei
- Project to: <sup>1</sup>Prefrontal cortical area & <sup>2</sup>cortical association areas
- Functions: <sup>1</sup>{Behavior & personality} + <sup>2</sup>{High intellectual cortical functions, e.g., speech}

➤ Thalamic syndrome (Déjerine–Roussy syndrome):

\*a/e: Cerebrovascular stroke (ischemia or hge), usually d.t. thrombosis of thalamo-geniculate artery (a branch of PCA) → damage of lat. & ventral nuclei, while ant. & med. nuclei remain intact

\*C/P: (on the opposite side of lesion)

1. Sensory: Hemianesthesia: Complete loss of all sensations, but facial sensations are usually intact (Why?)

- Epicritic sensations are permanently lost

- After few weeks or months, protopathic sensations are recovered, with return of pain

- Thalamic pain: has ↑↑ threshold (but once produced, it is felt maximum)

“Thalamic hyperpathia”

2. Motor: Hemiparesis or asthenia (ms. weakness)

3. Mixed hemiataxia: (Why?)

4. Emotional disturbances

\*ttt: Opiates + Anticonvulsants + Anti-depressants

## 2. Limbic System (LS)

➤ Components:

LS is a group of structures that make the border between the neocortex & the brainstem

(A) Limbic lobe (old cortex):

- Around corpus callosum

- Includes: \*Subcallosal gyrus \*Cingulate gyrus \*Retrosplenial cortex

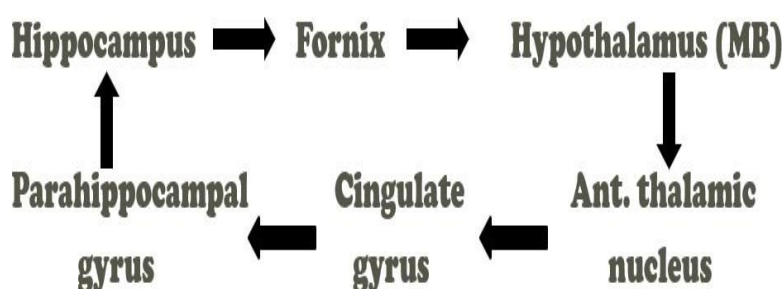
\*Entorhinal cortex \*Hippocampus \*Parahippocampal gyrus \*Uncus \*Pyriform cortex

### (B) Related subcortical structures:

- Include: \*Amygdala “in temporal lobe” \*Hypothalamus \*Ant. thalamic nucleus  
\*Septal nuclei \*Fornix \*Limbic midbrain area (LMA) “in upper part of midbrain”

### ➤ Connections:

1. Many interconnections (.) its different parts, especially (.) hypothalamus &:  
(a) Limbic lobe (b) Amygdala *via* stria terminalis (c) LMA
2. Few connections (.) LS & neocortex, especially (.) hypothalamus & orbitofrontal cortex
3. **Closed Papez circuit:** “Excitatory reverberating circuit” → After discharge



### ➤ Functions:

#### 1. **Olfaction:** (The oldest function of LS)

(a) Pyriform cortex: Perception of smell sensation

(b) Amygdala: \*Emotional response to olfactory stimuli (like or dislike)

\*Olfactory memories

*N.B.* Orbitofrontal cortex is responsible for conscious analysis of odors

## 2. Feeding behavior:

\*Amygdala: differentiates edible from inedible food

*N.B.* Damage of amygdala → Hyperphagia, omniphagia & hyperorality

## 3. Sexual behavior:

\*Amygdala: Instinctual sexual desire & innate sexual reactions that lead to mating

*N.B.* - Orbitofrontal cortex controls sexual behavior in humans

- Damage of amygdala → Hypersexuality (only in ♂)

## 4. Maternal behavior:

\*Cingulate & retrosplenial gyri: Nursing & care of babies

*N.B.* Prolactin & oxytocin facilitate maternal behavior & maternal bond

## 5. Emotions:

\*Definition: States of strong feelings, associated with autonomic & endocrinal changes

\*Involve:

(a) Cognition = To identify the stimulus (the cause)

(b) Affect = The feeling itself (+ve or -ve)

(c) Conation = The drive to take action

(d) Physical changes: e.g., ↑↑ HR, ↑↑ BP, sweating, etc.

\*Emotional reactions: Fear – Rage – Placidity

*N.B.* Fear & rage are protective reactions which are closely-related

|                  | <b>Fear</b>                                                                                                                                                                                                                                                   | <b>Rage</b>                                                                                                                                                                                                                                                                                 | <b>Placidity</b>                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>*Include:</b> | <p>Sense of insecurity d.t. impending danger or evil, associated with:</p> <p>-<u>Autonomic response:</u><br/>Sweating &amp; mydriasis</p> <p>-<u>Somatic response:</u><br/>Cowering, avoidance, turning head from side to side to seek escape &amp; flee</p> | <p>Violent anger, associated with:</p> <p>-<u>Autonomic response:</u><br/>Generalized symp. ++ (↑↑ HR, ↑↑ BP, mydriasis &amp; piloerection)</p> <p>-<u>Somatic response:</u><br/>Taking the attack position &amp; fighting (In cats: hissing, spitting, growling, biting &amp; clawing)</p> | <p>Extreme calmness with <u>no response</u> to provocation</p>               |
| <b>*Center:</b>  | <u>Fear center (FC)</u> in the periventricular nucleus of the hypothalamus                                                                                                                                                                                    | <u>Rage center (RC)</u> in the lat. nucleus of the hypothalamus                                                                                                                                                                                                                             | <u>Placidity center (PC)</u> in the ventromedial nucleus of the hypothalamus |
| <b>*Control:</b> | -Amygdala → ++ FC                                                                                                                                                                                                                                             | -Amygdala → ++ RC<br>-Neocortex & PC → -- RC                                                                                                                                                                                                                                                | -Amygdala → -- PC                                                            |

*N.B.* - Damage of amygdala → Placidity & disappearance of fear response

### - Sham rage reaction:

**\*Definition:** Exaggerated rage reaction in response to minor stimuli

**\*Causes:** 1. Lesion of neocortex      2. Lesion of ventromedial nucleus of hypothalamus (PC)

3. Stimulation of amygdala

**6. Control of autonomic functions:** e.g., changes in HR, BP & respiration

**7. Control of endocrinal functions:** especially release of CA & ACTH

**8. Motivation:**

\*Definition: It is the force that activates a certain behavior to achieve a certain goal

\*Control:

|                      | Reward system                                                                                                                                                    | Punishment system                                                                                                                                         |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>*Centers:</u>     |                                                                                                                                                                  |                                                                                                                                                           |
| (a) Mainly:          | <u>Reward center</u> in the lat. & ventromedial nuclei of the hypothalamus                                                                                       | <u>Punishment center</u> in the peri-ventricular nucleus of the hypothalamus                                                                              |
| (b) Others:          | Amygdala & LMA                                                                                                                                                   | Amygdala & hippocampus                                                                                                                                    |
| <u>*Stimulation:</u> | <u>Sense of reward</u> (relaxation, pleasure & satisfaction) →<br><u>Approach reaction:</u> the subject is motivated to approach or repeat the rewarding stimuli | <u>Sense of punishment</u> (displeasure, fear or terror) →<br><u>Avoidance reaction:</u> the subject is motivated to avoid or avert the punishing stimuli |

*N.B.* Dopamine is the transmitter in the reward system. Cocaine produces euphoria by ↑↑ the dopaminergic activity in the reward system

**9. Memory & learning:**

- This occurs via: \*Papez circuit      \*Motivation (++ of reward or punishment systems)

- Role of LS in memory:

(a) Hippocampus is the center of “memory encoding”. It sorts out the sensory signals into:

\*Significant ones which are projected to MB → ant. thalamic nucleus → prefrontal cortex → basal forebrain → neocortex (to be stored) & amygdala

\*Insignificant ones which are ignored

(b) Amygdala is responsible for the emotional responses to memories

- Role of LS in learning:

Amygdala & hippocampus are important for operant conditioning (see later)

➤ Klüver–Bucy syndrome:

\*a/e: Bilateral temporal lobectomy → damage of amygdaloid nuclei on both sides

\*C/P:

- |                                               |                               |
|-----------------------------------------------|-------------------------------|
| 1. Hyperphagia, omniphagia & hyperorality     | 2. Hypersexuality (only in ♂) |
| 3. Placidity & disappearance of fear response | 4. Depression of memory       |

### 3. Hypothalamus

It is a major central component of limbic system (LS)

➤ Site:

At the base of the brain, below & ant. to the thalamus, forming the floor & part of the wall of the 3<sup>rd</sup> ventricle

➤ Hypothalamic nuclei:

5 groups:

1. Ant. group: \*Preoptic n. \*Supraoptic n. \*Paraventricular n.  
\*Suprachiasmatic n. (SCN)

2. Post. group: \*Post. n. \*Mammillary body (MB)
3. Med. group: \*Ventromedial n. \*Dorsomedial n. \* Arcuate n.
4. Lat. group: \*One large lat. n.
5. Periventricular nucleus

➤ Nervous connections:

(A) Afferents from:

1. LS (limbic lobe, amygdala & hippocampus)
2. Thalamus (non-specific thalamic n.)
3. Basal ganglia (GP)
4. Reticular formation (collaterals from ascending sensory tracts)
5. Retina (optic tract → suprachiasmatic n.)

(B) Efferents to: (3 directions)

1. Upwards: → Ant. thalamic n. → Lim “closed circuit of Papez”
2. Downwards: → Reticular formation → Control centers in brainstem & spinal cord
3. → Post. pituitary gland (*via* hypothalamo-hypophyseal tract)

➤ Functions:

1. Autonomic control:

- a) Post. & lat. n. → ++ Symp. functions
- b) Ant. n. → ++ Parasymp. functions

2. Body temperature control: “Thermostat”

- a) Post. n. → Heat gain center

b) Ant. n. → Heat loss center

3. Correlation with sleep: (No direct role)

a) Fibers of ARAS pass through post. hypothalamus

b) Ant. hypothalamus: may contain a sleeping center

4. Diurnal (circadian) rhythm control:

*e.g.*, ACTH secretion, sleep-wake cycles & body temp. rhythm *via* retino-hypothalamic fibers to SCN

5. Endocrinal control:

a) Ant. pituitary gland: Releasing & inhibiting hormones *via* HHPC

b) Post. pituitary gland: ADH & oxytocin *via* HHT

c) Adrenal medulla: *via* adrenaline secreting center in MO

6. Food intake control: “Appestat”

a) Lat. n. → Feeding center

b) Ventromedial n. → Satiety center

7. General metabolism control: *via* 5 & 6

8. H<sub>2</sub>O balance:

a) H<sub>2</sub>O intake: Thirst center in lat. n.

b) H<sub>2</sub>O loss: ADH release from supraoptic & paraventricular n.

9. As a part of LS:

a) Control of motivation, memory & learning

b) Control of emotional reactions (fear, rage & placidity)

➤ **Effects of hypothalamic lesions:** (vary according to site of lesion)

1. Autonomic disturbances: e.g., spontaneous VC or VD, sweating, etc.
2. Body temp. disturbances: Hyper or hypothermia
3. Sleep disturbances:
  - a) Damage of post. hypoth. → Prolonged sleep & narcolepsy
  - b) Irritative lesion of post. hypoth. → Insomnia
4. Disturbed circadian rhythms
5. Endocrinal disorders: e.g., central DI & Fröhlich's syndrome
6. Food intake disturbances: Hyperphagia or anorexia

## Lecture (15): Ascending Reticular Activating System, Electroencephalography & Physiology of Sleep

**(Dr. Wagdy Nashaat Habeeb)**

### 1. Ascending Reticular Activating System (ARAS or RAS)

#### ➤ Definition:

It is a multi-neuronal polysynaptic system of nerve fibers

Origin: Facilitatory reticular formation (FRF)

Pathway: "Reticulo-thalamo-cortical"

Fibers from facilitatory RF extend upwards → non-specific thalamic nuclei → all areas of CC

*N.B.* Some fibers project directly to CC without relay in thalamus